SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

CO1 ASSESSMENT TOOL 1 – Case Study Analysis

Course Code:	CSA10 / CSA1063	**Course Title:**	Software Engineering
CO Assessed:	CO1	**Assessment:**	Assessment Tool 1 – Case Study Analysis
Weightage:	20%	**Total Marks:**	25
CO1 Statement:	*Apply advanced methodologies and agile project management techniques to manage software projects effectively*		
Student Name:	Shaik Mohammad Sohan	**Reg. No.:**	192525042
Date:		**Duration:**	60 minutes

Code of Conduct

I 192525042 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student:

Annexure A – Case Study Analysis

Instructions to Students:

Each student will be allotted **one problem statement**. Analyze the assigned problem systematically by

following the steps given below. Your report should demonstrate your understanding of software engineering concepts and your ability to design an appropriate software solution.

Based on your allotted problem statement, prepare a report by answering the following questions:

1. ### Understand the Problem Statement
   - Explain the problem in your own words.
   - Identify the objectives of the proposed system.
   - State the scope of the problem.
2. ### Identify Key Stakeholders and Challenges
   - Identify all the stakeholders involved in the system.
   - Explain the roles and responsibilities of each stakeholder.
   - Discuss the major challenges and issues faced by the stakeholders.

3. ## Analyze the Current Approach

   - Describe the existing system or current process.
   - Identify its strengths and limitations.
   - Analyze the problems or gaps that need to be addressed.

4. ## Propose a Suitable Solution Using Software Engineering Principles

   - Design a software-based solution for the given problem.
   - Describe the major functional modules of the proposed system.
   - Identify the functional and non-functional requirements.
   - Explain how software engineering principles such as modularity, scalability, maintainability, reliability, usability, and security are incorporated into your solution.

5. Justify the Chosen Methodology/Framework

- Select an appropriate Software Development Life Cycle (SDLC) model or software development framework (e.g., Agile, Scrum, Waterfall, Spiral, DevOps).
- Justify why the selected methodology/framework is suitable for the proposed system.
- Explain how it supports successful project development and maintenance.

6. Discuss Expected Outcomes and Limitations

- Describe the expected benefits of the proposed solution.
- Explain how the solution addresses the identified challenges.
- Discuss potential risks, implementation challenges, and limitations.
- Suggest possible future enhancements.

Rubrics for Calculation

Criteria	Excellent (5 Marks)	Good (3 Marks)	Satisfactory (2 Marks)	Needs Improvement (1 Mark)
Problem Understanding	Clearly explains the problem, objectives, and scope with complete understanding.	Explains the problem with minor omissions.	Basic understanding with limited explanation.	Poor understanding; objectives and scope are unclear or missing.
Stakeholder & Challenge Analysis	Identifies all stakeholders, their roles, and challenges with clear analysis.	Identifies most stakeholders and challenges with adequate analysis.	Identifies only a few stakeholders or challenges.	Stakeholders and system analysis are incomplete or inaccurate.
Current System Analysis	Thoroughly analyzes the existing system, strengths, weaknesses, and identifies all gaps.	Good analysis with most strengths and weaknesses identified.	Limited analysis with partial identification of issues.	Proposed solution is unclear or inappropriate; requirements are largely missing.

Proposed Software Solution & Software Engineering Principles	Proposes an innovative, feasible solution applying appropriate software engineering principles (modularity, scalability, maintainability, security, etc.).	Suitable solution with good application of software engineering principles.	Basic solution with limited application of software engineering principles.	Methodology is inappropriate, poorly justified, or not discussed.

Methodology/ Framework Justification	Selects the most suitable SDLC/model/framework and provides strong technical justification.	Suitable methodology selected with adequate justification.	Methodology selected but justification is weak.	Outcomes, limitations, or future enhancements are missing; report lacks organization and clarity.

Criteria	**Mark**
Problem Understanding	/5
Stakeholder & Challenge Analysis	/5
Current System Analysis	/5
Proposed Software Solution & Software Engineering Principles	/5
Methodology/ Framework Justification	/5
TOTAL	/5

CHAPTER 1: INTRODUCTION

1.1 Background of the Problem

Food waste has become one of the most serious global challenges affecting both society and the environment. Every day, restaurants, supermarkets, hotels, and event organizers prepare more food than they can sell or serve. A significant amount of this surplus food is still safe and suitable for consumption, but it is often thrown away due to the lack of an organized redistribution system. At the same time, many charities, orphanages, shelters, and underprivileged communities struggle to provide enough food for people in need. The gap between food donors and charities results in unnecessary hunger and wastage of valuable resources. Manual communication methods such as phone calls and personal contacts are slow and unreliable, causing food to expire before it reaches beneficiaries. Additionally, food waste contributes to environmental pollution by increasing greenhouse gas emissions from landfills. A digital platform that connects food donors with registered charities can help solve this problem by enabling quick and efficient food redistribution. Such a solution ensures that edible food reaches needy people instead of being wasted.

1.2 Importance of Solving the Problem

Solving the problem of food waste is essential because it benefits society, the environment, and the economy. Every meal saved can help reduce hunger among vulnerable communities and improve food security. By redistributing surplus food instead of discarding it, organizations can reduce waste disposal costs and use resources more efficiently. Reducing food waste also helps decrease methane emissions from landfills, contributing to environmental protection and climate change mitigation. A software-based solution improves coordination between food donors and charities through real-time communication and automated notifications. It increases transparency by maintaining accurate records of donations and food distribution. The platform also encourages businesses to participate in social responsibility initiatives and strengthen their public image. Furthermore, efficient food redistribution reduces pressure on charitable organizations and improves their ability to serve more people. Overall, solving this problem creates a sustainable system that benefits both current and future generations.

1.3 Related SDG Goal(s)

This case study primarily supports **United Nations Sustainable Development Goal (SDG) 12: Responsible Consumption and Production**, which aims to reduce waste and promote the efficient use of natural resources. One of the targets of SDG 12 is to significantly reduce global food waste at the retail and consumer levels. The proposed platform also contributes to **SDG 2:**

Zero Hunger by ensuring that surplus food reaches people who lack access to nutritious meals. In addition, it supports **SDG 13: Climate Action** by reducing greenhouse gas emissions caused by decomposing food waste in landfills. Through improved food redistribution, the system promotes sustainable consumption and responsible production practices. The platform encourages collaboration between businesses, nonprofit organizations, and communities to achieve common sustainability goals. By reducing unnecessary waste and supporting vulnerable populations, the software contributes to long-term social and environmental.

1.4 Objectives of the Case Study

The primary objective of this case study is to design a software platform that efficiently connects food donors with registered charities for timely food redistribution. The proposed system aims to reduce food waste by enabling donors to share surplus food before it expires. It seeks to improve communication between restaurants, supermarkets, event organizers, and charitable organizations through a centralized digital platform. Another objective is to automate donation requests, approvals, and pickup scheduling to reduce manual work and increase operational efficiency. The system also aims to provide real-time notifications and GPS-based location services to ensure quick food collection. Maintaining transparency through donation tracking, reports, and user verification is another important objective. The case study also focuses on developing a secure, scalable, and user-friendly software solution using software engineering principles. Finally, the project aims to support sustainable development by reducing hunger, protecting the environment, and promoting responsible consumption practices in accordance with the United Nations Sustainable Development Goals.

CHAPTER 2: PROBLEM IDENTIFICATION & ANALYSIS

2.1 Problem Description

Many restaurants, supermarkets, and event organizers prepare more food than they eventually sell or use. Since there is no efficient platform to connect food donors with charities, much of this food is discarded despite being safe to eat. Charitable organizations often face difficulty finding available food donations quickly because communication is done manually through phone calls or personal contacts. This process is slow, inefficient, and lacks proper coordination. As a result, edible food is wasted while many people remain hungry. The absence of real-time information, tracking, and notifications makes food redistribution difficult.

2.2 Objectives

- Reduce food wastage through timely redistribution.
- Connect food donors with registered charities.
- Automate food donation requests and approvals.
- Provide real-time notifications.
- Improve transparency in donation activities.
- Reduce manual communication.
- Track donation history and reports.
- Increase operational efficiency.
- Support sustainable development.
- Improve user satisfaction.

2.3 Scope

- User registration and login.
- Food donation posting.
- Charity request management.
- Food pickup scheduling.
- GPS-based nearby charity search.
- Notification system.

- Admin monitoring.
- Reports and analytics.

Out of Scope

- Online food selling.
- Payment gateway integration.
- International food donation.
- Medical assistance.
- Grocery shopping services.
- Restaurant inventory management.
- Delivery by third-party logistics.
- AI-based food quality testing.

CHAPTER 3: STAKEHOLDER ANALYSIS

Stakeholder	**Role**
Administrator	Manages users, verifies charities, monitors donations, generates reports, and controls the system.
Restaurant Owner	Uploads surplus food details and manages donations.
Supermarket Manager	Donates unsold but edible food.
Event Organizer	Shares leftover food after events.
Charity Organization	Requests available food and arranges pickup.
Volunteers	Assist in food collection and delivery.

Stakeholder	Role
Government	Ensures food safety regulations and policy compliance.
Food Safety Officer	Verifies food quality standards.
System Developer	Maintains and updates the software.
Database Administrator	Manages system database and security.

CHAPTER 4: EXISTING SYSTEM ANALYSIS

4.1 Existing Process

Currently, food donation mainly depends on manual communication through phone calls, WhatsApp messages, emails, or personal contacts. Restaurants contact charities individually whenever surplus food is available. Many charities are unaware of available donations because there is no centralized platform. Communication delays often result in food spoilage before collection. There is no proper tracking, reporting, or verification of donations. Records are maintained manually or not recorded at all. This process is inefficient and lacks transparency. Donors cannot easily identify nearby charities, and charities cannot quickly locate available food. Consequently, significant quantities of edible food are wasted every day.

4.2 Strengths

- Simple process.
- No software investment required.
- Direct communication.
- Easy for small organizations.
- Minimal technical knowledge needed.
- Flexible coordination.
- Immediate contact possible.
- Low maintenance cost.

4.3 Weaknesses

- Time-consuming.
- No centralized database.
- Lack of transparency.
- Food often expires before pickup.
- Difficult tracking.
- Poor communication.
- Manual record keeping.
- Limited scalability.

4.4 Gap Analysis

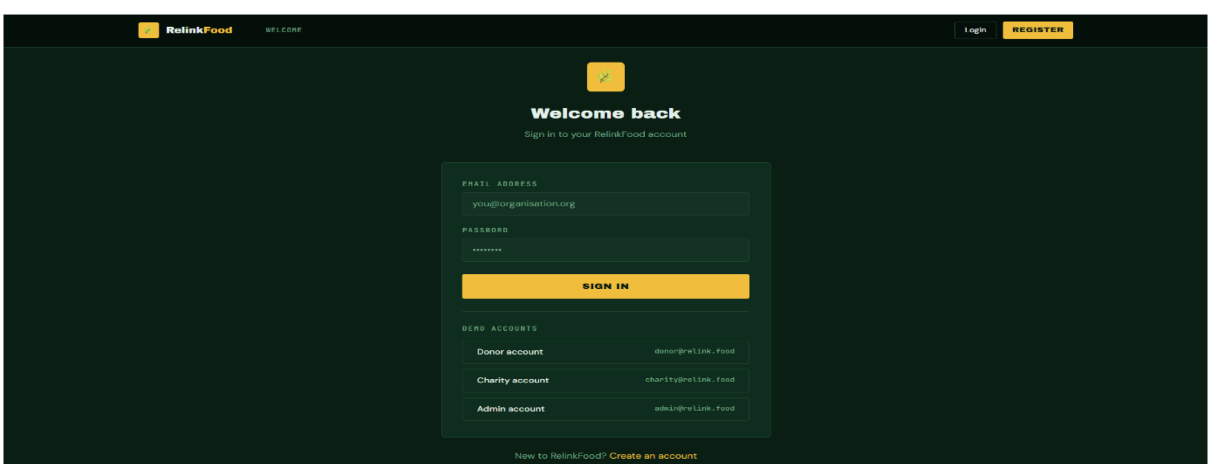
The existing system does not provide real-time food availability, automated notifications, donation tracking, user verification, GPS-based matching, or centralized management. There is no reporting system, leading to poor accountability and reduced efficiency. A software platform is required to eliminate these gaps and improve food redistribution.

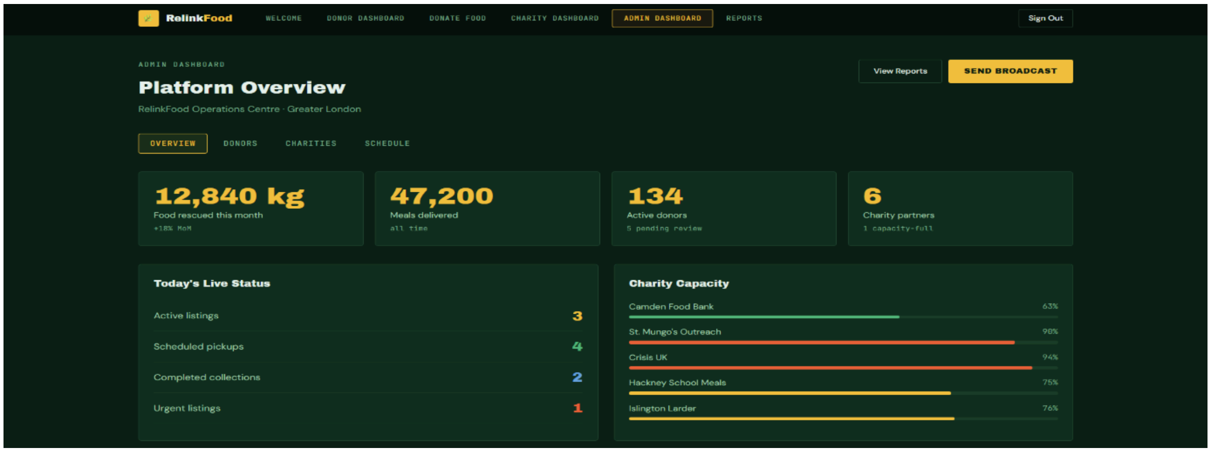
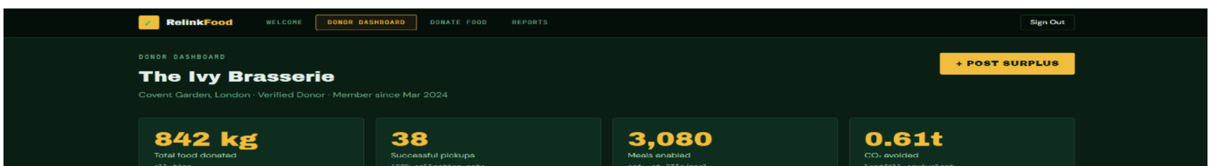
CHAPTER 5: PROPOSED SOFTWARE SOLUTION

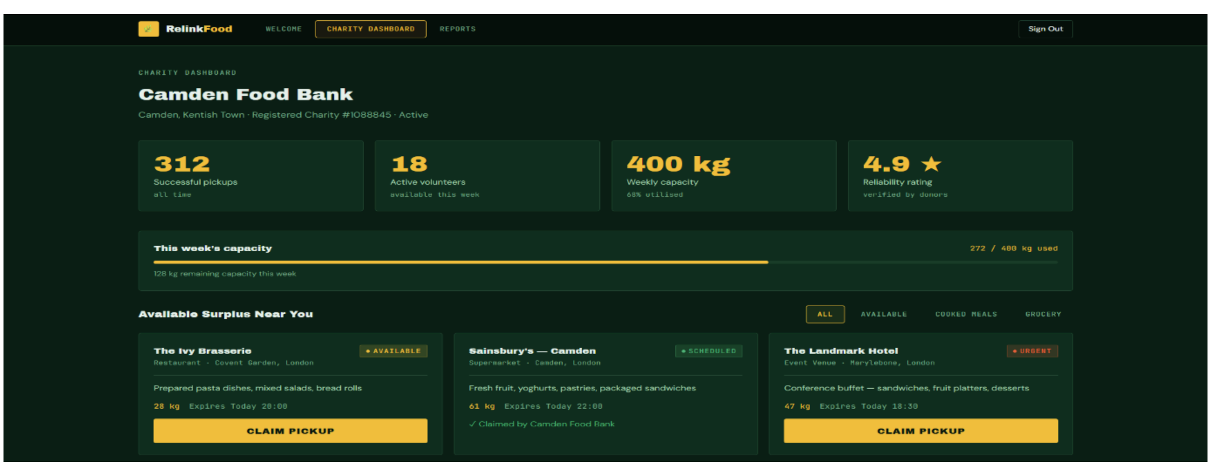
5.1 Proposed System

The proposed solution is a web and mobile-based Sustainable Food Waste Reduction Platform that connects restaurants, supermarkets, and event organizers with registered charities. Donors can upload surplus food details, including quantity, expiry time, and location. Nearby charities receive instant notifications and can request the food. The administrator verifies users and monitors all activities. The system provides GPS location services, reports, dashboards, and donation tracking. Automatic notifications ensure timely collection before food expires. The software improves transparency, reduces food waste, and supports sustainable food management.

5.2 Functional Modules




RelinkFood
WELCOME
DONOR DASHBOARD
DONATE FOOD
CHARITY DASHBOARD
ADMIN DASHBOARD
REPORTS
Sign Out
ADMIN DASHBOARD
Platform Overview
RelinkFood Operations Centre · Greater London
View Reports
SEND BROADCAST
OVERVIEW
DONORS
CHARITIES
SCHEDULE
12,840 kg
Food rescued this month
+18% MoM
47,200
Meals delivered
all time
134
Active donors
5 pending review
6
Charity partners
1 capacity-full
Today's Live Status
Active listings 3
Scheduled pickups 4
Completed collections 2
Urgent listings 1
Charity Capacity
Camden Food Bank 63%
St. Mungo's Outreach 90%
Crisis UK 94%
Hackney School Meals 75%
Islington Larder 76%

RelinkFood
WELCOME
DONOR DASHBOARD
DONATE FOOD
REPORTS
Sign Out
DONOR DASHBOARD
The Ivy Brasserie
Covent Garden, London · Verified Donor · Member since Mar 2024
+ POST SURPLUS
842 kg
Total food donated
all time
38
Successful pickups
100% collection rate
3,080
Meals enabled
est. at 273g/meal
0.61t
CO₂ avoided
landfill equivalent

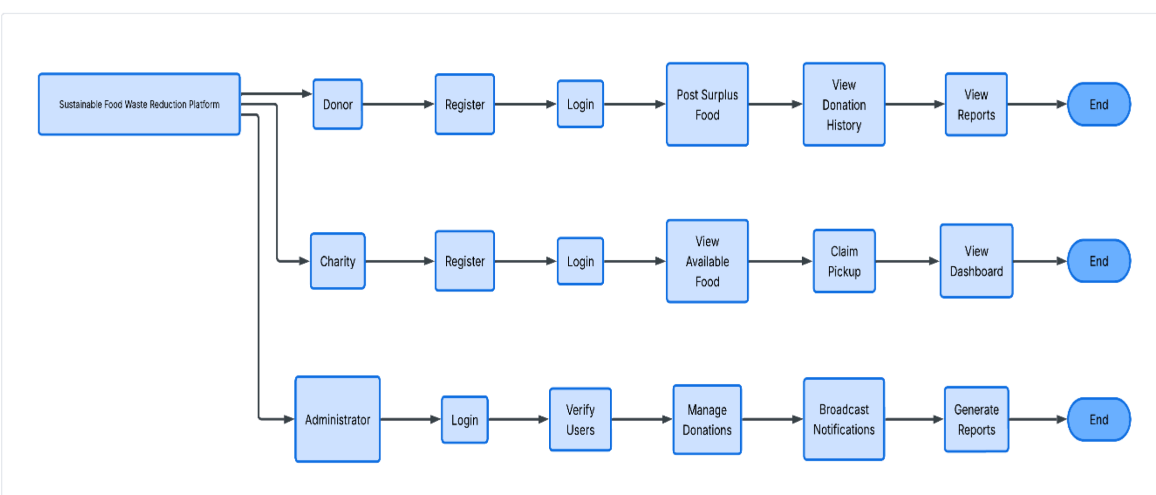
RelinkFood
WELCOME
DONOR DASHBOARD
DONATE FOOD
REPORTS
Sign Out
IMPACT REPORTS
The Ivy Brasserie — Impact Report
Greater London · 2026
WEEK
MONTH
QUARTER
YEAR
Download PDF
12,840 kg
Food rescued
this month
47,200
Meals delivered
est. at 273g/serving
8.2t
CO₂ avoided
vs. landfill baseline
£34,600
Estimated food value
redistributed this month
Monthly Food Rescued (kg)
Jan – Jul 2026
820
1,040
960
1,380
1,520
1,740
Jan
Feb
Mar
Apr
May
Jun
Jul

RelinkFood
WELCOME
CHARITY DASHBOARD
REPORTS
Sign Out
CHARITY DASHBOARD
Camden Food Bank
Camden, Kentish Town · Registered Charity #1088845 · Active
312
Successful pickups
all time
18
Active volunteers
available this week
400 kg
Weekly capacity
68% utilised
4.9 ★
Reliability rating
verified by donors
This week's capacity
272 / 400 kg used
128 kg remaining capacity this week
Available Surplus Near You
ALL
AVAILABLE
COOKED MEALS
GROCERY
The Ivy Brasserie
● AVAILABLE
Restaurant · Covent Garden, London
Prepared pasta dishes, mixed salads, bread rolls
28 kg Expires Today 20:00
CLAIM PICKUP
Sainsbury's — Camden
● SCHEDULED
Supermarket · Camden, London
Fresh fruit, yoghurts, pastries, packaged sandwiches
61 kg Expires Today 22:00
✓ Claimed by Camden Food Bank
The Landmark Hotel
● URGENT
Event Venue · Marylebone, London
Conference buffet — sandwiches, fruit platters, desserts
47 kg Expires Today 18:30
CLAIM PICKUP

5.3 Software Engineering Principles Applied

The proposed system follows modular architecture by separating different functionalities into independent modules such as registration, donation management, notifications, reporting, and administration. The software is scalable, allowing more users and charities to join without affecting performance. Security is maintained through authentication, authorization, and encrypted data storage. Reliability is achieved through regular backups and error handling mechanisms. Maintainability is ensured by writing clean and modular code that can be updated easily. Reusable components reduce development time for future enhancements. A simple user interface improves usability for both donors and charities. Flexible design enables future integration with mobile applications, AI, cloud computing, and IoT technologies.

CHAPTER 6: SYSTEM DESIGN

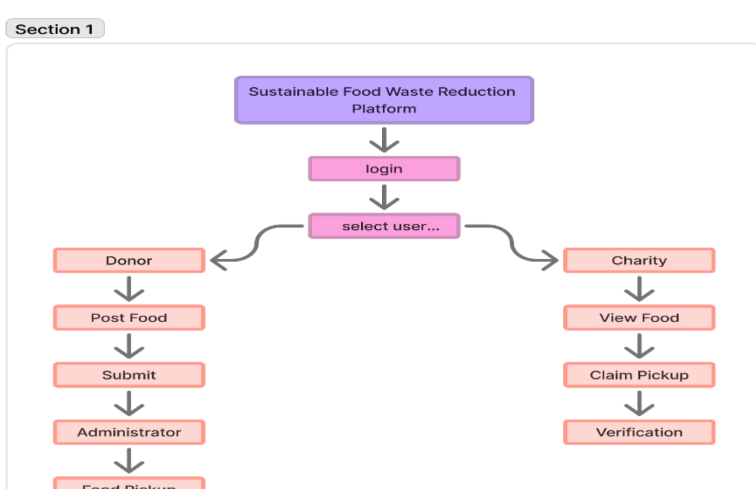
6.1 Use Case Diagram



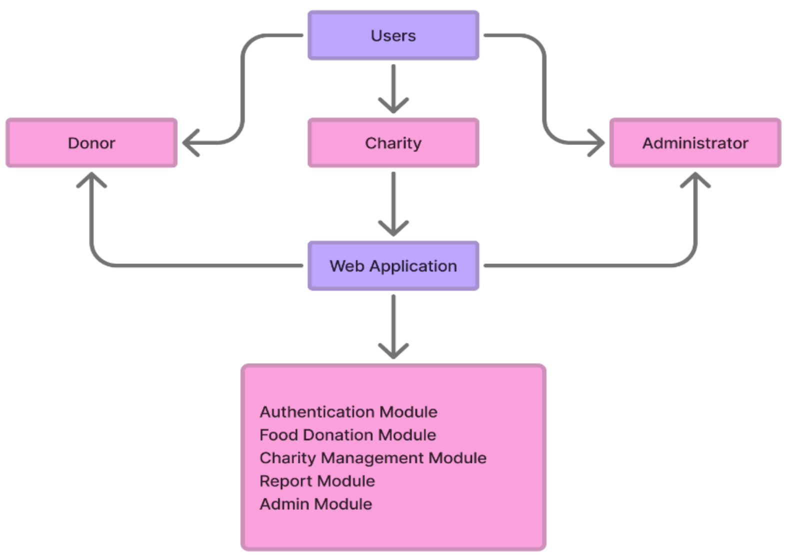
6.2 Use Case Description

Use Case	Actor	Description
Register	Donor/Charity	Creates a new account.
Login	All Users	Authenticates users.
Post Surplus Food	Donor	Uploads food details.
View Available Food	Charity	Displays nearby donations.
Claim Pickup	Charity	Requests available food.
Verify Users	Admin	Approves charities and donors.
Generate Reports	Admin	Produces donation reports.
Broadcast Notification	Admin	Sends alerts to all users.

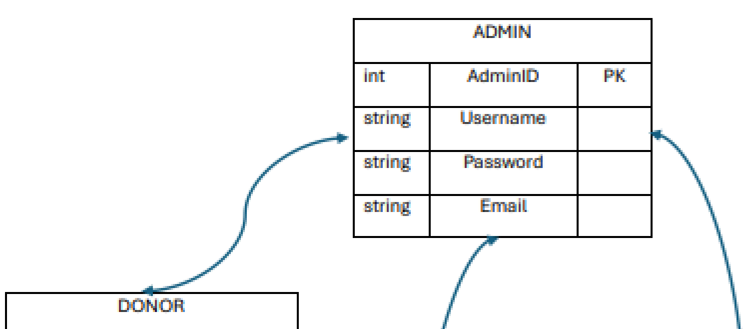
6.3 Activity Diagram



6.4 System Architecture Diagram



6.5 Database ER Diagram



CHAPTER 7: METHODOLOGY / SDLC SELECTION

7.1 Selected Model – Agile SDLC

The **Agile Software Development Life Cycle (SDLC)** model is selected for the Sustainable Food Waste Reduction Platform because it supports continuous development and frequent improvements based on user feedback. Agile divides the project into small iterations called sprints, allowing developers to build, test, and improve different modules step by step. This approach enables the development team to quickly respond to changing requirements from restaurants, charities, and administrators. Regular communication between stakeholders ensures that the software meets user expectations and maintains high quality. Since the platform requires features such as real-time notifications, dashboards, and reports, Agile provides flexibility for adding new features without affecting the entire system. It also encourages teamwork, faster delivery, and continuous testing throughout the development process. Therefore, Agile is the most suitable SDLC model for this project.

Why Agile Model?

- Flexible to changing requirements.
- Faster software development.
- Continuous testing and improvement.
- Better communication with stakeholders.
- Early delivery of working software.
- Improved software quality.
- Easy to add new features.
- High customer satisfaction.

Advantages

- Faster development cycle.
- Reduced project risk.
- Continuous customer feedback.
- Better collaboration.
- Easy maintenance.
- Early bug detection.
- Higher productivity.
- Improved software quality.

Suitability

Agile is suitable because the project involves multiple users such as donors, charities, and administrators whose requirements may change during development. The platform also requires frequent updates, testing, and improvements to ensure smooth food donation and redistribution. Agile allows the team to release modules such as login, donation management, and reporting in stages while continuously improving the system based on user feedback.

Development Stages

1. Requirement Gathering
2. System Analysis
3. System Design
4. Development
5. Testing
6. Deployment
7. Maintenance

CHAPTER 8: IMPLEMENTATION PLAN

The implementation of the Sustainable Food Waste Reduction Platform will follow a planned schedule to ensure successful development and deployment. During the first phase, project requirements will be collected from restaurants, charities, and administrators. After analyzing the requirements, the system architecture, database, and user interface will be designed. Developers will then implement modules such as registration, login, food donation, request management, notifications, and reports. Once development is completed, the software will undergo testing to identify and fix any errors. After successful testing, the system will be deployed for real-world use. Finally, regular maintenance and updates will be provided to improve performance and add new features.

CHAPTER 9: EXPECTED OUTCOMES

The proposed platform is expected to significantly reduce food waste by connecting food donors with registered charities in real time. Restaurants, supermarkets, and event organizers will be able to donate surplus food quickly, while charities can easily locate and request available food. The system will improve operational efficiency through automation, reduce manual communication, and provide accurate reports for monitoring donation activities. Businesses will strengthen their corporate social responsibility by participating in food donation programs. The platform will help vulnerable communities receive nutritious food while reducing environmental pollution caused by food waste. Overall, the project supports sustainable resource management and contributes to multiple United Nations Sustainable Development Goals.

Benefits

- Reduces food waste.
- Faster food redistribution.
- Saves operational costs.
- Improves transparency.
- Better communication.
- Easy donation tracking.

- Efficient management.
- Supports sustainability.

Business Impact

- Improves company reputation.
- Enhances corporate social responsibility.
- Reduces disposal expenses.
- Better resource utilization.

User Benefits

- Easy registration.
- Quick food donation.
- Real-time notifications.
- Simple dashboards.
- Secure platform.

Social Impact

- Reduces hunger.
- Helps charities.
- Protects the environment.
- Supports local communities.

SDG Contribution

- SDG 12 – Responsible Consumption and Production.
- SDG 2 – Zero Hunger.
- SDG 13 – Climate Action.

CHAPTER 10: RISKS AND LIMITATIONS

The proposed software may face several risks during implementation and operation. The platform depends on a stable internet connection for real-time communication and notifications. Data privacy and cybersecurity threats must be addressed through secure authentication and encrypted storage. Limited project budgets may delay development and future upgrades. Some users may hesitate to adopt the system because of limited technical knowledge or resistance to change. Incorrect food information or delayed pickup may reduce the effectiveness of the donation process. Server failures or software bugs may temporarily affect system availability. Regular maintenance, user training, and strong security measures are necessary to minimize these risks and ensure reliable operation.

Possible Risks

- Internet dependency.

- Data privacy issues.
- Cybersecurity threats.
- Budget limitations.
- Server downtime.
- Technical failures.

CHAPTER 11: FUTURE ENHANCEMENTS

The platform can be enhanced by integrating advanced technologies to improve efficiency and user experience. Artificial Intelligence can predict surplus food availability and recommend suitable charities automatically. A chatbot can provide instant customer support and answer user queries. Predictive analytics can forecast food donation trends and optimize distribution. IoT sensors can monitor food temperature during transportation to ensure food safety. A mobile application will improve accessibility for donors and charities. Blockchain technology can provide secure and transparent donation records. Cloud deployment can improve system scalability, storage, and performance for future expansion.

Future Features

- AI-based food prediction.
- Chatbot support.
- Predictive analytics.
- IoT monitoring.
- Mobile application.
- Blockchain security.
- Cloud deployment.
- GPS route optimization.

CHAPTER 12: CONCLUSION

The Sustainable Food Waste Reduction Platform offers an effective solution for reducing food waste and helping communities in need. The proposed software connects restaurants, supermarkets, event organizers, and charities through a centralized digital platform that simplifies food donation and redistribution. Features such as user registration, food posting, notifications, dashboards, and reports improve efficiency, transparency, and accountability. The system benefits donors, charities, administrators, and society by reducing hunger, minimizing environmental pollution, and promoting responsible resource management. The project successfully applies software engineering principles and supports the United Nations Sustainable Development Goals. Overall, the platform demonstrates how technology can solve real-world social problems while creating long-term sustainable benefits. In conclusion, the Sustainable Food Waste Reduction Platform demonstrates how software engineering can be applied to solve a real-world social and environmental challenge. The proposed system benefits donors, charities, administrators, and society by improving food redistribution, reducing waste, promoting sustainability, and enhancing transparency. With future enhancements such as Artificial Intelligence, IoT, cloud computing, blockchain, and mobile applications, the platform has the potential to become a comprehensive and scalable solution for sustainable food management, creating long-term positive impacts for both people and the environment.

CHAPTER 13: REFERENCES

1. **Varghese, C., Pathak, D., & Varde, A. S. (2021)** – Developed the **SeVa** mobile application for connecting food donors with people in need to reduce food waste.
2. **Damiani, M., Pastorello, T., Carlesso, A., et al. (2021)** – Explained that redistributing surplus food is more sustainable than disposing of it.
3. **Oroski, F. A., & Silva, J. M. (2023)** – Reviewed digital platforms and technologies used for food waste reduction and food donation.
4. **IEEE (2023)** – Published research on digital technologies that improve food waste management and food supply chain efficiency.
5. **Food and Agriculture Organization (FAO) (2024)** – Highlighted the importance of reducing food loss and food waste to improve global food security.
6. **Kafa, N., Jaegler, A., & Hong, J. (2024)** – Studied the adoption of mobile applications for food waste reduction and redistribution.
7. **United Nations Environment Programme (UNEP) (2024)** – Emphasized food waste reduction as an important step toward achieving **SDG 12: Responsible Consumption and Production**.
8. **IEEE – ICISS Conference (2025)** – Proposed Artificial Intelligence techniques to reduce food waste across the food supply chain.
9. **Procedia Computer Science (2025)** – Presented a smart food donation platform using real-time location and notification services.
10. **IEEE – ICRTEECT Conference (2025)** – Introduced AI-based forecasting methods to minimize restaurant food waste and improve sustainability.

CHAPTER 14: APPENDIX

A. Survey Questions

- How often do you have surplus food?
- Do you currently donate surplus food?
- What challenges do you face during food donation?
- Would you use an online donation platform?
- How important are real-time notifications?

B. Screenshots

- Login Page
- Donor Dashboard
- Charity Dashboard
- Admin Dashboard
- Reports Dashboard

C. Additional Diagrams

- Use Case Diagram
- Activity Diagram
- ER Diagram
- System Architecture Diagram
- Workflow Diagram

D. Sample Reports

- Donation Report
- Food Collection Report
- Monthly Impact Report
- Charity Activity Report
- Admin Dashboard Report

WEB-BASED APPLICATION LINK

https://sky-linter-56105931.figma.site/